COMPUTER PROCESSOR

BY: ENGR. GLA

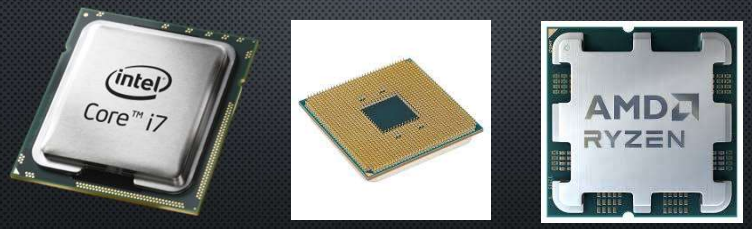
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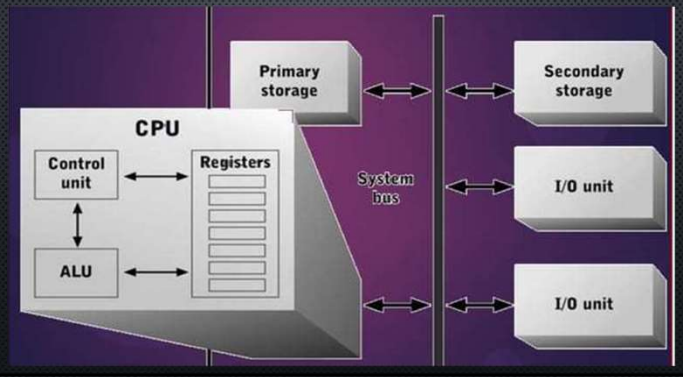
INTRODUCTION

- A PROCESSOR OR CPU IS AN ELECTRONIC CIRCUIT THAT CAN EXECUTE COMPUTER PROGRAMS
- A PROCESSOR IS THE LOGIC CIRCUITRY THAT RESPONDS TO AND PROCESSES THE BASIC INSTRUCTIONS THAT DRIVES A COMPUTER
- A COMPUTER ANALYZES DATA AND CONTROLS DATA FLOW IN A COMPUTER
- IT HANDLES THE CENTRAL MANAGEMENT FUNCTIONS OF A HIGH-POWERED PC

MICROPROCESSOR IMAGE



REVIEW THE COMPONENTS OF COMPUTER

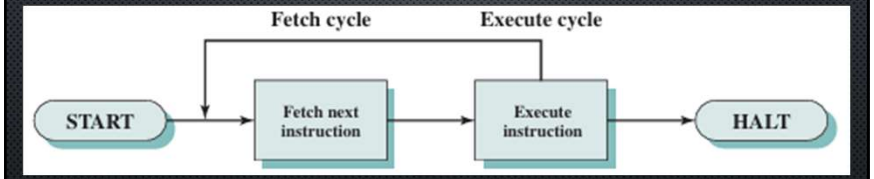


WORKING OF PROCESSOR

THE WORKING OF PROCESSOR MAINLY CONSISTS OF FOUR STEPS:

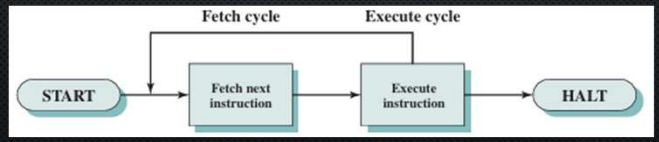
1. **FETCH** – RETRIEVES PROGRAM INSTRUCTIONS FROM MEMORY
2. **DECODE** – INSTRUCTIONS ARE BROKEN DOWN INTO PARTS
3. **EXECUTE** – CPU PERFORMS THE OPERATION IMPLIED BY THE PROGRAM INSTRUCTION
4. **WRITE BACK** – PROCESSOR WRITES BACK THE RESULTS OF EXECUTION TO THE COMPUTER'S MEMORY

BASIC INSTRUCTION CYCLE



THE FETCH INSTRUCTION

1. THE FETCHED INSTRUCTION IS LOADED INTO A REGISTER IN THE PROCESSOR KNOWN AS THE INSTRUCTION REGISTER (IR).
2. THE INSTRUCTION CONTAINS BITS THAT SPECIFY THE ACTION THE PROCESSOR IS TO TAKE.
3. THE PROCESSOR INTERPRETS THE INSTRUCTION AND PERFORMS THE REQUIRED ACTION.

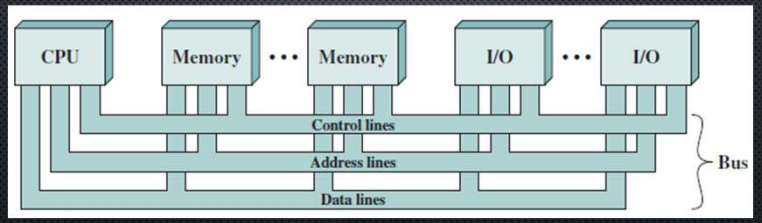


THE FETCH INSTRUCTION

IN GENERAL, THESE ACTIONS FALL INTO FOUR CATEGORIES:

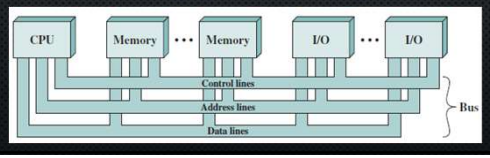
- **PROCESSOR-MEMORY:** DATA MAY BE TRANSFERRED FROM PROCESSOR TO MEMORY OR FROM MEMORY TO PROCESSOR.
- **PROCESSOR-I/O:** DATA MAY BE TRANSFERRED TO OR FROM A PERIPHERAL DEVICE BY TRANSFERRING BETWEEN THE PROCESSOR AND AN I/O MODULE.
- **DATA PROCESSING:** THE PROCESSOR MAY PERFORM SOME ARITHMETIC OR LOGIC OPERATION ON DATA.
- **CONTROL:** AN INSTRUCTION MAY SPECIFY THAT THE SEQUENCE OF EXECUTION BE ALTERED. FOR EXAMPLE, THE PROCESSOR MAY FETCH AN INSTRUCTION FROM LOCATION 149, WHICH SPECIFIES THAT THE NEXT INSTRUCTION BE FROM LOCATION 182. THE PROCESSOR WILL REMEMBER THIS FACT BY SETTING THE PROGRAM COUNTER TO 182. THUS, ON THE NEXT FETCH CYCLE, THE INSTRUCTION WILL BE FETCHED FROM LOCATION 182 RATHER THAN 150

BUS INTERCONNECTION



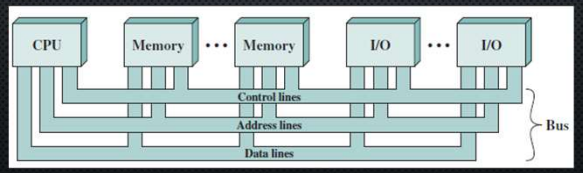
BUS INTERCONNECTION

The **data lines** provide a path for moving data among system modules. These lines, collectively, are called the **data bus**. The data bus may consist of 32, 64, 128, or even more separate lines, the number of lines being referred to as the **width** of the data bus Because each line can carry only 1 bit at a time, the number of lines determines how many bits can be transferred at a time. The width of the data bus is a key factor in determining overall system performance.



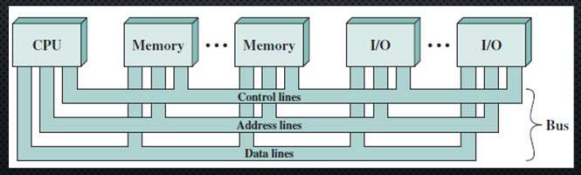
BUS INTERCONNECTION

The **address lines** are used to designate the source or destination of the data on the data bus. For example, if the processor wishes to read a word (8, 16, or 32 bits) of data from memory, it puts the address of the desired word on the address lines. Clearly, the width of the **address bus** determines the maximum possible memory capacity of the system.



BUS INTERCONNECTION

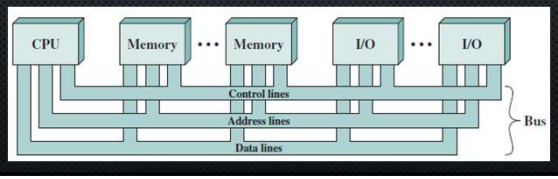
The **control lines** are used to control the access to and the use of the data and address lines. Because the data and address lines are shared by all components, there must be a means of controlling their use.



BUS OPERATION

If one module wishes to send data to another, it must do two things:
(1) obtain the use of the bus, and
(2) transfer data via the bus.

If one module wishes to request data from another module, it must:
(1) obtain the use of the bus, and
(2) transfer a request to the other module over the appropriate address lines/data lines. It must then wait for that second module to send the data



BASIC KNOWLEDGE ABOUT PROCESSOR

CLOCK SPEED

IS A MEASURE OF HOW QUICKLY A COMPUTER COMPLETES BASIC COMPUTATIONS AND OPERATIONS. IT IS MEASURED AS A FREQUENCY IN HERTS.

CORE

HARDWARE UNIT IN THE PROCESSOR ARCHITECTURE THAT CAN EXECUTE INSTRUCTIONS SENT TO IT.

HYPER THREADING

THREAD ARE THE VIRTUAL CORE AND WORK LIKE A REAL CORE INSIDE CPU. WHEN CORES ASSEMBLED WITH MULTI THREADS THEN IT IS KNOWN AS HYPER THREADING TECHNOLOGY.

BASIC KNOWLEDGE ABOUT PROCESSOR

CACHE

FIRST BLOCK OF RAM WHICH INTERACT BETWEEN THE MAIN MEMORY AND CPU USING CACHE CONTROLLER CHIP.

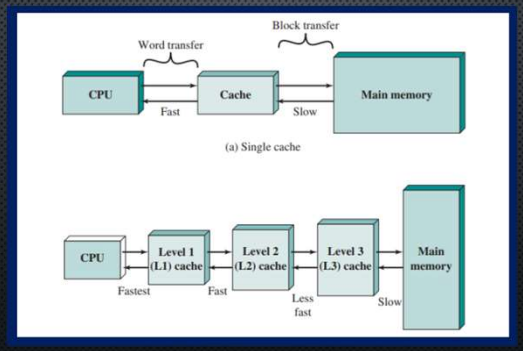
MULTI PROCESSING

A PROCESSING TECHNIQUE IN WHICH MULTIPLE PROCESSORS OR MULTIPLE PROCESSING CORES IN A SINGLE COMPUTER EACH WORK ON A DIFFERENT JOB.

PARALLEL PROCESSING

A PROCESSING TECHNIQUE IN WHICH MULTIPLE PROCESSORS OR MULTIPLE PROCESSING CORES IN A SINGLE COMPUTER WORK TOGETHER TO COMPLETE ONE JOB MORE QUICKLY.

CACHE MEMORY



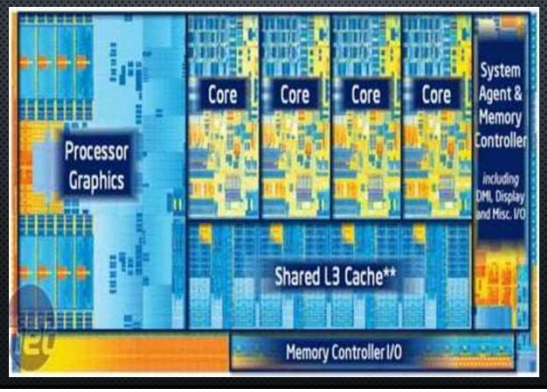
SINGLE, DOUBLE, AND MULTI CORE PROCESSORS

Single core: Has one core to process different operations like intel Pentium.

Dual core : Has two cores to process operations; able to process more information at the same time compare to single core like intel core i3 and i5.

Quad core: Contains two dual core processors in one integrated circuit and generally used for multi tasking like intel core i7.

INTEL I7 PROCESSOR



POPULAR BRANDS OF PROCESSOR

Intel : The Intel computer processor is exclusively designed by Intel. Its latest and most popular models include Intel hyper thread technology that speed up processor speed .

AMD : The AMD computer processor is exclusively made by Advanced Micro Devices, Inc. (AMD). It provides excellent performance and value. It is compatible with most off-the-shelf computer programs and applications. Some AMD Computer Processors are programmed with built-in anti-virus protection

POPULAR BRANDS OF PROCESSOR





ACTIVITY 1

1. Search for the history of processors.
2. What general categories of functions are specified by computer instructions?
3. What are the future expectations about processors?
4. Explain the operation of the basic instruction cycle based on the discussion.